

EduGenie AI

01 – Brainstorming and Ideation

Project Idea: EduGenie is an AI-powered educational assistant designed to help students learn concepts in a simple and understandable way.

1. Problem Identification

Students may find it difficult to understand complex concepts, revise long texts, create quizzes, and prepare a structured learning plan. They often need simple explanations and guided practice.

2. Proposed Solution

EduGenie provides an interactive educational assistant that can answer student questions, explain concepts, summarize text, generate quizzes, and create personalized learning paths.

3. Target Users

- College students
- Beginner learners
- Students preparing for examinations
- Learners who need simple explanations and practice

4. Main Features

- Ask educational questions
- Explain concepts in simple language
- Summarize educational text
- Generate multiple-choice quizzes
- Create personalized learning paths

5. Technology Ideas

- Python
- FastAPI
- Google Gemini API
- HTML, CSS and JavaScript
- Pydantic for data validation

6. Expected Benefits

EduGenie aims to make learning easier, faster, and more organized by providing clear explanations, revision support, practice questions, and structured learning plans.

7. Future Enhancement Ideas

- Student progress tracking

- More personalized recommendations
- Additional quiz types
- Learning history and saved content
- Support for more educational subjects

8. Conclusion

The brainstorming stage identified the main educational problems and defined EduGenie as a practical AI-based solution. These ideas form the foundation for the next project phases.